

Problem1

Relation Schemas	Meaning and Additional Information	Candidate key	Foreign key
Musicians(ssn: string, name: string, annualIncome: real) Primary Key: ssn	Registers the ssn of musicians and their names and annual incomes.	ssn (same as Primary Key)	None
Instruments(instrID: string, iname: string, musickey: string) Primary Key: instrID	Registers instruments	instrID (same as Primary Key)	None
Plays(ssn: string, instrID: string) Primary Key: (ssn, instrID)	Registers which musicians play which instruments.	(ssn, instrID) (same as Primary Key)	ssn → Musicians(ssn), instrID → Instruments(instrID)
SongsAppears(songID: string, authorSSN: string, title: string, albumIdentifier: string) Primary Key: songID	Registers songs and which albums they appear. Every song must have exactly one author who is a musician and appears in exactly one album.	songID (same as Primary Key)	authorSSN → Musicians(ssn), albumIdentifier → AlbumProducer(albumIdentifier)
Lives(ssn: string, aid: string, phone: string) Primary Key: (ssn, aid)	Registers where musicians live and which phones they use.	(ssn, aid) (same as Primary Key)	ssn → Musicians(ssn), aid → Place(aid)
Place(aid: string, address: string, otherInfo: string) Primary Key: aid	Register all addresses.	aid (same as Primary Key)	None
Perform(songID: string, ssn: string) Primary Key: (songID, ssn)	Register which songs are performed by which musicians.	(songID, ssn) (same as Primary Key)	songID → SongsAppears(songID), ssn → Musicians(ssn)
AlbumProducer(albumIdentifier: string, ssn: string, copyrightDate: date, speed: real, title: string) Primary Key: albumIdentifier	Registers albums and their producers, who are also musicians.	albumIdentifier (same as Primary Key)	ssn → Musicians(ssn)

Problem2

Assumptions:

Each song is written by exactly one musician.

Each song appears in exactly one album.

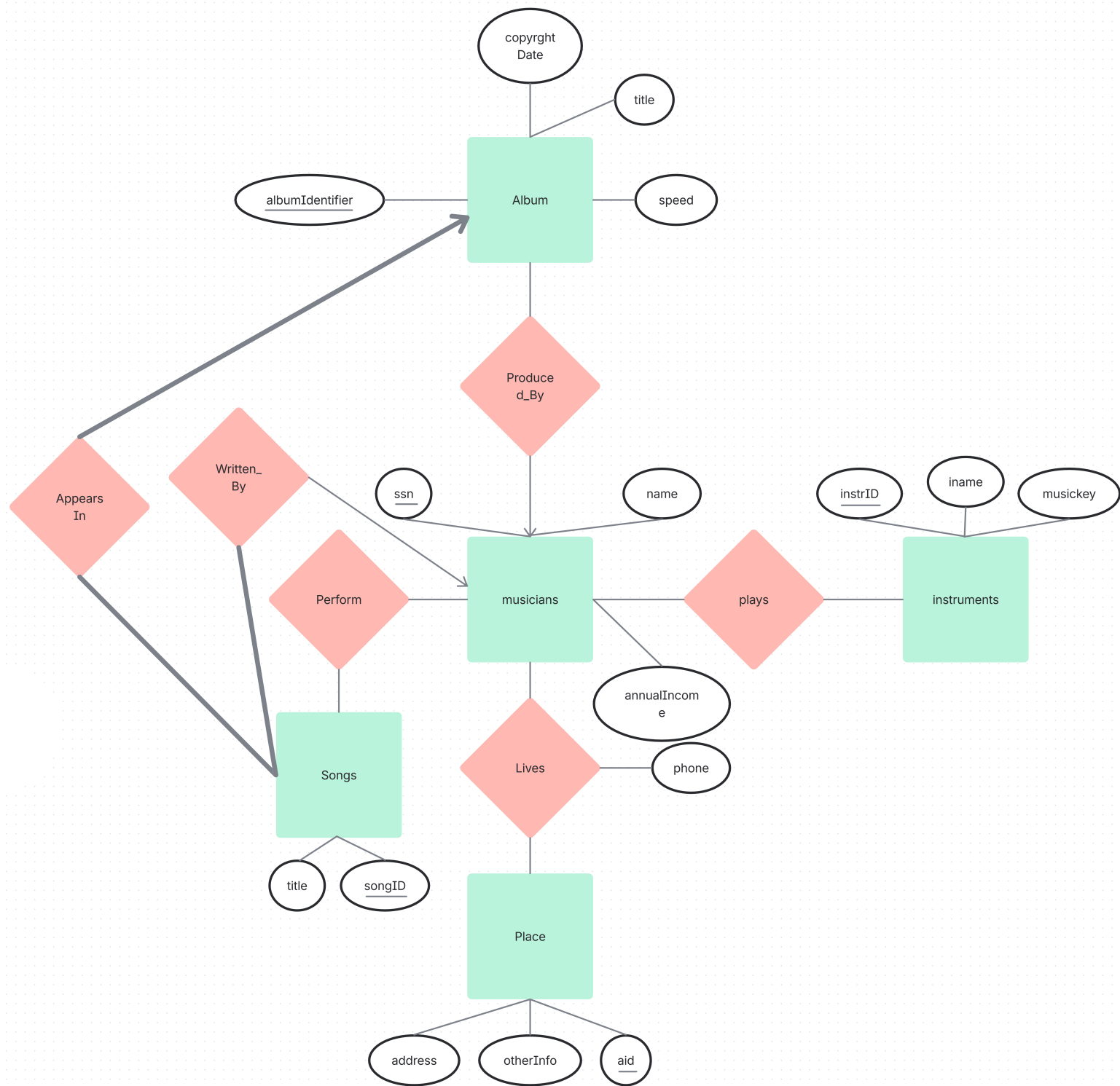
Each album must have at least one song.

A musician may perform many songs, and a song may be performed by many musicians.

Each album is produced by exactly one musician.

A musician may produce many albums.

Songs, albums, musicians, instruments, and places are strong entities.



Problem3

(1)

$\rho(\text{Temp1}, \pi_{\text{instrID}}(\sigma_{\{\text{iname}='Saxophone'\}}(\rho(I, \text{Instruments}))))$
 $\text{Result} \leftarrow \pi_{\text{ssn}, \text{name}}(\rho(M, \text{Musicians}) \bowtie (\rho(P, \text{Plays}) \bowtie \text{Temp1}))$

(2)

$\rho(T1, \rho(L, \text{Lives}) \bowtie \rho(P, \text{Place}))$
 $\rho(T2, T1 \bowtie \rho(M, \text{Musicians}))$
 $\rho(M\text{Addr}, \pi_{\text{ssn}, \text{name}, \text{address}}(T2))$
 $\rho(A, M\text{Addr})$
 $\rho(B, M\text{Addr})$
 $\text{Result} \leftarrow \pi_{A.\text{ssn}, A.\text{name}}(\sigma_{\{A.\text{address} = B.\text{address} \wedge A.\text{ssn} \neq B.\text{ssn}\}}(A \times B))$

(3)

$\rho(\text{Ben}, \sigma_{\{\text{name}='Ben'\}}(\rho(M, \text{Musicians})))$
 $\rho(T1, \text{Ben} \bowtie \rho(AP, \text{AlbumProducer}))$
 $\text{Result} \leftarrow \pi_{\text{albumIdentifier}, \text{title}}(T1)$

(4)

$\rho(\text{Ben}, \sigma_{\{\text{name}='Ben'\}}(\rho(M, \text{Musicians})))$
 $\rho(\text{BenPlay}, \text{Ben} \bowtie \rho(P, \text{Plays}))$
 $\rho(\text{BenInstr}, \pi_{\text{instrID}}(\text{BenPlay}))$
 $\rho(\text{AllBen}, \rho(P2, \text{Plays})/\text{BenInstr})$
 $\text{Result} \leftarrow \pi_{\text{ssn}, \text{name}}(\text{AllBen} \bowtie \rho(M2, \text{Musicians}))$

(5)

$\rho(G1, \sigma_{\{\text{iname}='guitar'\}}(\rho(I1, \text{Instruments})))$
 $\rho(G2, G1 \bowtie \rho(P1, \text{Plays}))$
 $\rho(G\text{ssn}, \pi_{\text{ssn}}(G2))$
 $\rho(Pi1, \sigma_{\{\text{iname}='piano'\}}(\rho(I2, \text{Instruments})))$
 $\rho(Pi2, Pi1 \bowtie \rho(P2, \text{Plays}))$
 $\rho(P\text{ssn}, \pi_{\text{ssn}}(Pi2))$
 $\rho(\text{Both}, G\text{ssn} \cap P\text{ssn})$
 $\rho(\text{Temp}, \text{Both} \bowtie \rho(AP, \text{AlbumProducer}))$
 $\text{Result} \leftarrow \pi_{\text{title}}(\text{Temp})$

(6)

$\rho(S544\text{Perf}, \sigma_{\{\text{songID}='544'\}}(\rho(PF, \text{Perform})))$
 $\rho(\text{PerfSSN}, \pi_{\text{ssn}}(S544\text{Perf}))$
 $\rho(S544\text{Song}, \sigma_{\{\text{songID}='544'\}}(\rho(SA, \text{SongsAppears})))$
 $\rho(\text{Alb544}, \pi_{\text{albumIdentifier}}(S544\text{Song}))$
 $\rho(\text{Prod544}, \text{Alb544} \bowtie \rho(AP, \text{AlbumProducer}))$
 $\rho(\text{ProdSSN}, \pi_{\text{ssn}}(\text{Prod544}))$

Result $\leftarrow \pi_{\text{ssn}, \text{name}}(\rho(\text{M}, \text{Musicians}) \bowtie (\text{PerfSSN} \cup \text{ProdSSN}))$

(7)

$\rho(\text{A}, \rho(\text{AP1}, \text{AlbumProducer}))$

$\rho(\text{B}, \rho(\text{AP2}, \text{AlbumProducer}))$

$\rho(\text{NotMax}, \pi_{\text{A.albumIdentifier}, \text{A.title}}(\sigma_{\{\text{A.speed} < \text{B.speed}\}}(\text{A} \times \text{B})))$

$\rho(\text{AllAlb}, \pi_{\text{albumIdentifier}, \text{title}}(\rho(\text{AP}, \text{AlbumProducer})))$

Result $\leftarrow \text{AllAlb} - \text{NotMax}$

(8)

$\rho(\text{AllSongs}, \pi_{\text{songID}, \text{title}}(\rho(\text{SA}, \text{SongsAppears})))$

$\rho(\text{PerfIDs}, \pi_{\text{songID}}(\rho(\text{PF}, \text{Perform})))$

$\rho(\text{AllIDs}, \pi_{\text{songID}}(\rho(\text{SA2}, \text{SongsAppears})))$

$\rho(\text{NotPerfIDs}, \text{AllIDs} - \text{PerfIDs})$

$\rho(\text{Ans}, \text{NotPerfIDs} \bowtie \rho(\text{SA3}, \text{SongsAppears}))$

Result $\leftarrow \pi_{\text{songID}, \text{title}}(\text{Ans})$

(9)

$\rho(\text{A}, \rho(\text{AP1}, \text{AlbumProducer}))$

$\rho(\text{B}, \rho(\text{AP2}, \text{AlbumProducer}))$

$\rho(\text{TwoAlb}, \pi_{\text{A.ssn}}(\sigma_{\{\text{A.ssn} = \text{B.ssn} \wedge \text{A.albumIdentifier} \neq \text{B.albumIdentifier}\}}(\text{A} \times \text{B})))$

$\rho(\text{SongsByThem}, \text{TwoAlb} \bowtie \rho(\text{PF}, \text{Perform}))$

$\rho(\text{Ans}, \text{SongsByThem} \bowtie \pi_{\{\text{TwoAlb.ssn} = \text{PF.ssn}\}}(\rho(\text{SA}, \text{SongsAppears})))$

Result $\leftarrow \pi_{\text{songID}, \text{title}}(\text{Ans})$

(10)

$\rho(\text{High}, \sigma_{\{\text{annualIncome} > 50000\}}(\rho(\text{M}, \text{Musicians})))$

$\rho(\text{ProdSSN}, \pi_{\text{ssn}}(\rho(\text{AP}, \text{AlbumProducer})))$

$\rho(\text{HighSSN}, \pi_{\text{ssn}}(\text{High}))$

$\rho(\text{TargetSSN}, \text{HighSSN} - \text{ProdSSN})$

$\rho(\text{Ans}, \text{TargetSSN} \bowtie \rho(\text{M2}, \text{Musicians}))$

Result $\leftarrow \pi_{\text{ssn}, \text{name}}(\text{Ans})$